Selection on Observables III: Propensity Scores, Weighting, Regression

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Roadmap

1. Theory: Potential outcomes, identification, key quantities
2. Randomization
   - difference in means, variance estimation
   - covariate adjustment
   - blocking
3. Selection on Observables
   - sub-classification
   - matching (on X)
   - weighting (on X)
   - matching and weighting on $Pr(D = 1)$ (propensity scores)
   - regression
4. Instrumental Variables
5. Regression Discontinuity
6. Difference in Differences and Synthetic Control
7. Sensitivity and Bounds

We have to keep saying this...

Assumption: Conditional Ignorability

$$\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp D_i \mid X_i = x \quad \text{for any} \quad x \in \mathcal{X}$$

(a.k.a. exogeneity, unconfoundedness, selection on observables, no omitted variables)

So we have "experiments" within each stratum of X, but how to estimate an average effect as if we did it strata-wise then averaged? ("conditioning on X")

- We took very literal approach with sub-classification and matching
- Now, let's think about re-weighting in ways that make the overall distribution of treated and control more similar
- This will allow us to estimate things as if we have done so conditionally on X then averaged over some density in X

Propensity Scores: The conventional motivation

- May be tough to get close matches on X_i when multi-dimensional
- What if you could just match on a one-dimensional summary?

Conditioning on the Propensity Score

Under SOO and common support,

$$\{Y_i(0), Y_i(1)\} \perp\!\!\!\perp D_i \mid \pi(X_i)$$

The intuition:

- Among those with $\pi(X_i) = c$, you have $Pr(D_i = 1) = c$ no matter what differences in X_i you see.
- That is, D_i does not depend upon X_i, conditionally on $\pi(X_i)$.
- Once D_i no longer depends on X_i, it also no longer depends on $\{Y_{1i}, Y_{0i}\}$ (SOO).
- So its like random experiments within each stratum of $\pi(X_i)$.

Identification with Propensity Scores

Proof.

Show that $\Pr(D = 1 | Y_0, Y_1, \pi(X)) = \Pr(D = 1 | \pi(X)) = \pi(X)$, implying independence of (Y_0, Y_1) and D conditional on $\pi(X)$.

$$\begin{aligned}
\Pr(D = 1 | Y_1, Y_0, \pi(X)) &= \mathbb{E}[D | Y_1, Y_0, \pi(X)] \\
&= \mathbb{E}\left[\mathbb{E}[D | Y_1, Y_0, X] | Y_1, Y_0, \pi(X)\right] \text{ (LIE)} \\
&= \mathbb{E}\left[\mathbb{E}[D | X] | Y_1, Y_0, \pi(X)\right] \text{ (SOO)} \\
&= \mathbb{E}\left[\pi(X) | Y_1, Y_0, \pi(X)\right] \\
&= \pi(X)
\end{aligned}$$

Using a similar argument

$$\begin{aligned}
\Pr(D = 1 | \pi(X)) &= \mathbb{E}[D | \pi(X)] = \mathbb{E}[\mathbb{E}[D | X] | \pi(x)] \\
&= \mathbb{E}[\pi(X) | \pi(X)] = \pi(X)
\end{aligned}$$

therefore $\Pr(D = 1 | Y_1, Y_0, \pi(X)) = \Pr(D = 1 | \pi(X))$ □

Using Propensity Scores

First, estimate $P(D_i = 1|X_i) = \hat{\pi}_i$, e.g. by logit. Then condition on it:

- Matching:
  - may need a *caliper* to trim cases without common support
  - or, trim cases with extreme pscores by hand
  - but beware, this changes your estimand
- Weighting
  - Simple IPW weights: $\frac{1}{p(D_i|X_i)} = \frac{1}{D_i\pi_i+(1-D_i)(1-\pi_i)}$
  - Stabilized IPW weights: $\frac{p(D_i)}{p(D_i|X_i)}$
  - But if using weights, check how extreme they get and how few units are doing most of the work

Using Propensity Scores

Pros:

- Just one thing to match on
- Ignores imbalance on X's that do not predict D...very helpful for large P
- Will give you balance ("balancing property"), but *only in expectation* and if correctly estimated

Cons:

- If balancing property seen as the *goal* of pscores – by that metric it often under-performs. (See King & Nielsen 2015)
- Requires correct model and estimation of the propensity score!
- Extreme weights can cause large bias (so check)

IPW weights that also guarantee balance?

- Covariate balancing propensity scores (CBPS; Imai, Ratkovic 2014)
- For continuous treatments, CBGPS (Fong, Hazlett, Imai 2017)

Using Propensity Scores: Blattman & Annan

```
pi.out = glm(abd~age+fthr_ed+mthr_ed+hh_size96+orphan96+C_ach+C_akw+
C_ata+C_kma+C_oro+C_pad+C_paj+C_pal,data=dat,family="binomial"(link=logit)

matchout.pi=Match(Y=dat$educ,Tr=dat$abd,X=pi.out$fit,estimand="ATT")
summary(matchout.pi)

#Check balance
mb.out.pi =MatchBalance(match.out=matchout.pi,abd~age+fthr_ed+
mthr_ed+hh_size96+orphan96+C_ach+C_akw+C_ata+C_kma+
C_oro+C_pad+C_paj+C_pal,data=dat)

btest_after_pi=baltest.collect(mb.out.pi,var.names=varnames,after=TRUE)

balancecompare.pi=cbind(btest[,c("mean.Tr","mean.Co","T pval",
"KS pval")],btest_after_pi[,c("mean.Tr","mean.Co","T pval","KS pval")])
```

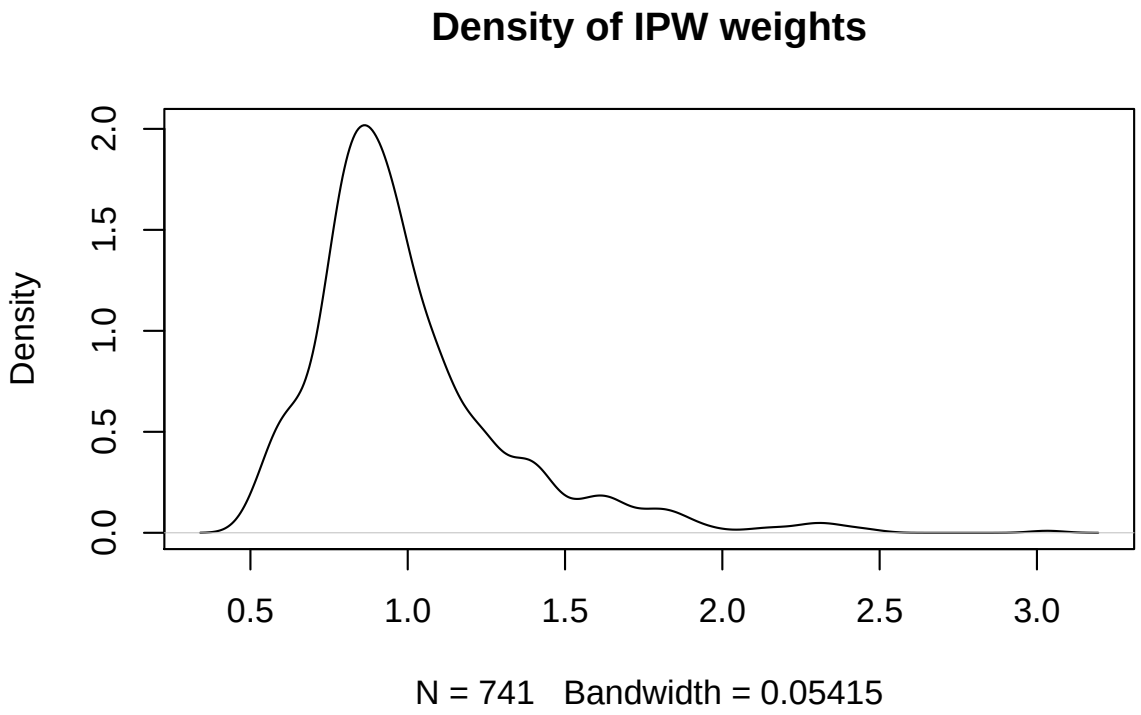
Using Propensity Scores: Blattman & Annan

```
 balancecompare.pi
          mean.Tr mean.Co T pval KS pval mean.Tr mean.Co T pval KS pval
age         21.37   20.15   0.00    0.00   21.37   21.04   0.27    0.17
fthr_ed      5.76    6.07   0.27    0.87    5.76    5.60   0.48    0.09
mthr_ed      2.09    2.49   0.09    0.35    2.09    2.40   0.11    0.20
hh_size96    8.09    8.70   0.06    0.04    8.09    8.02   0.79    0.03
orphan96     0.08    0.08   0.90      NA    0.08    0.11   0.07      NA
C_ach        0.15    0.11   0.13      NA    0.15    0.17   0.60      NA
C_akw        0.16    0.08   0.00      NA    0.16    0.17   0.62      NA
C_ata        0.10    0.20   0.00      NA    0.10    0.09   0.55      NA
C_kma        0.15    0.12   0.19      NA    0.15    0.16   0.87      NA
C_oro        0.05    0.14   0.00      NA    0.05    0.05   0.95      NA
C_pad        0.12    0.12   0.98      NA    0.12    0.09   0.10      NA
C_paj        0.15    0.10   0.06      NA    0.15    0.15   0.91      NA
C_pal        0.11    0.13   0.51      NA    0.11    0.13   0.45      NA
```

Using Propensity Scores: Blattman & Annan

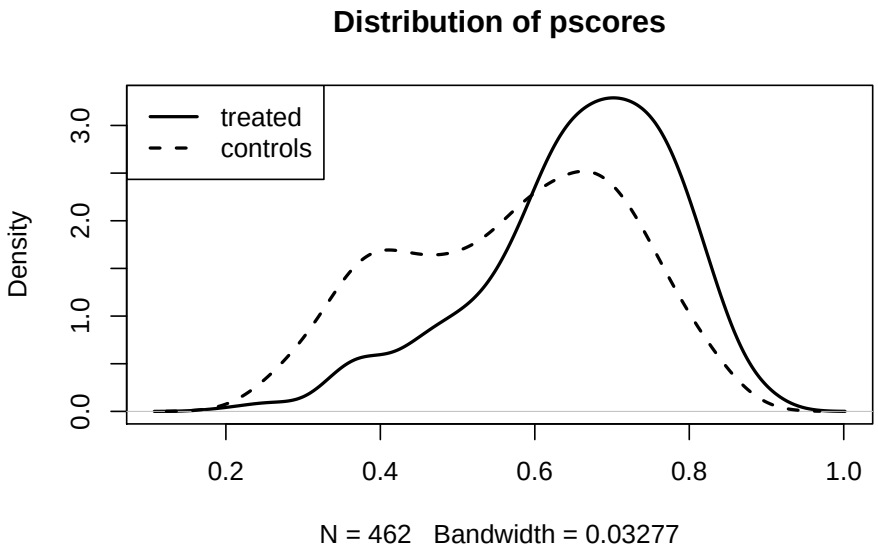
Weighting on the propensity scores (stabilized IPW):

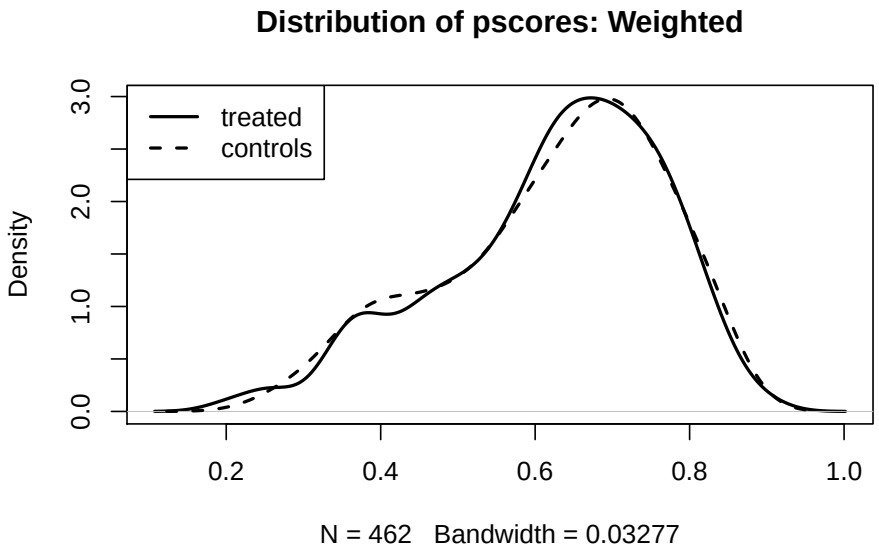
```
ps=pi.out$fit
D=dat$abd
PrD=mean(D)
IPW= (D*PrD+(1-D)*(1-PrD))/(D*ps+(1-D)*(1-ps))

#Good to check how crazy the weights are:
plot(density(IPW))
```



How does $p(\pi_i)$ look before/after weighting?

```
plot(density(pi.out$fit[D==1], weight=IPW[D==1]/sum(IPW[D==1])))
lines(density(pi.out$fit[D==0], weight=IPW[D==0]/sum(IPW[D==0])),lwd=2)
legend("topleft", legend=c("treated","controls"), lty=c(1,2))
```





Check Omnibus Balance

```
omnibus.bal = lm(abd~age+fthr_ed+mthr_ed+hh_size96+orphan96+
                   C_ach+C_akw+C_ata+C_kma+C_oro+C_pad+C_paj+C_pal, data=dat)
summary(omnibus.bal)
#R2=0.08, F-statistic: 5.585 on 12 and 728 DF,  p-value: 3.303e-09

omnibus.bal.ipw = lm(abd~age+fthr_ed+mthr_ed+hh_size96+orphan96+
                   C_ach+C_akw+C_ata+C_kma+C_oro+C_pad+C_paj+C_pal,
                   weight=IPW, data=dat)
summary(omnibus.bal.ipw)

Coefficients: (1 not defined because of singularities)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  6.138e-01  1.058e-01   5.804 9.69e-09 ***
age          5.474e-04  3.691e-03   0.148    0.882
fthr_ed     -4.053e-04  5.352e-03  -0.076    0.940
mthr_ed     -2.116e-04  6.628e-03  -0.032    0.975
hh_size96    4.875e-04  4.689e-03   0.104    0.917
orphan96     3.979e-03  6.873e-02   0.058    0.954
---
Multiple R-squared:  0.0002747,Adjusted R-squared:  -0.0162
F-statistic: 0.01667 on 12 and 728 DF,  p-value: 1
```

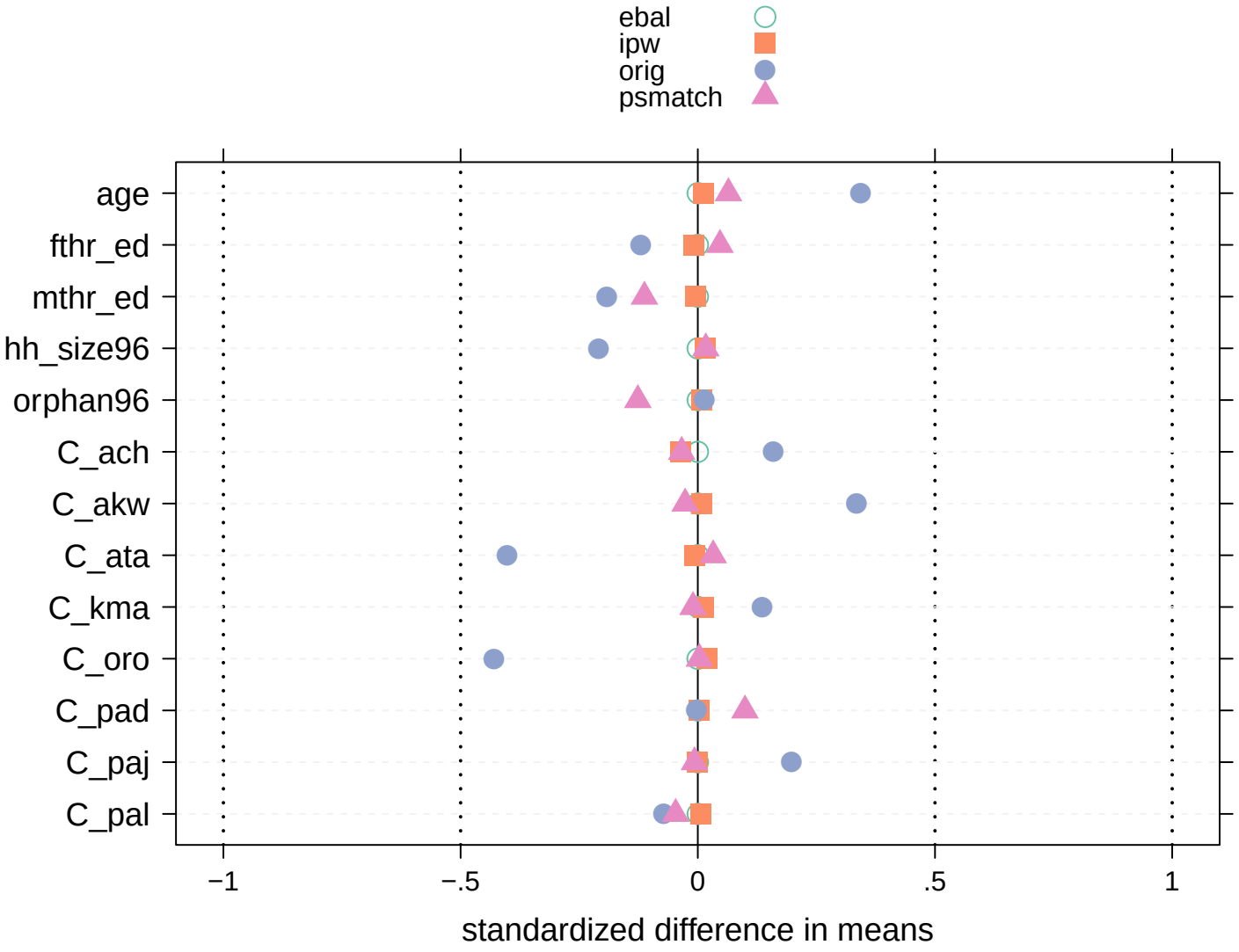
Yay!

Estimating the ATE with IPW weights

```
lm.out.naive = lm(educ~abd, data=dat)
lm.out.ipw = lm(educ~abd, weight=IPW, data=dat)
lm.out.ipw2=lm(educ~abd+age+fthr_ed+mthr_ed+hh_size96+orphan96+
    C_ach+C_akw+C_ata+C_kma+C_oro+C_pad+C_paj+C_pal,weight=IPW,data=dat)
```

	Dependent variable:		
	educ		
	(1)	(2)	(3)
abd	-0.595^{***}	-0.726^{***}	-0.735^{***}
	(0.218)	(0.220)	(0.210)
Constant	7.416^{***}	7.499^{***}	6.113^{***}
	(0.172)	(0.174)	(0.610)
Observations	741	741	741
R^2	0.010	0.014	0.128
Adjusted R^2	0.009	0.013	0.112
Residual Std. Error (df = 739)	2.876	2.904	2.754
Note:		$^{*}p<0.1$; $^{**}p<0.05$; $^{***}p<0.01$	

Balance Plot after IPW



A different view on weighting

For my money I prefer to think of weights as follows:

- We have conditional treatment effects in each stratum of X:

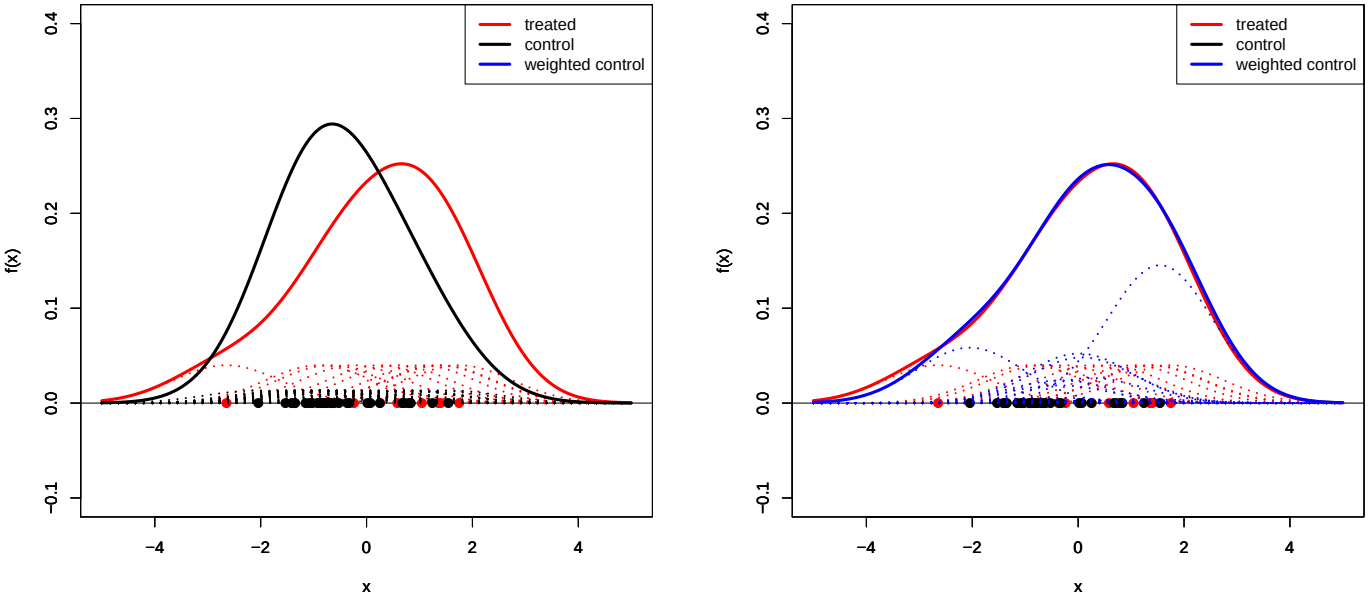
$$\mathbb{E}[\tau|X] = \mathbb{E}[Y_i|D_i = 1, X_i] - \mathbb{E}[Y_i|D_i = 0, X_i]$$

- We wish we could take this $\mathbb{E}[\tau|X]$ and average it up over some distribution, say $p(X)$:

$$\begin{aligned}\mathbb{E}_X\mathbb{E}[\tau|X] &= \int \mathbb{E}[Y_i|D_i = 1, X_i] - \mathbb{E}[Y_i|D_i = 0, X_i]p(X)dx \\ &= \int \mathbb{E}[Y_i|D_i = 1, X_i]p(X)dx - \int \mathbb{E}[Y_i|D_i = 0, X_i]p(X)dx \\ &= \text{mean for treated minus controls when same density, p(X)}\end{aligned}$$

Preview: we could get this if we could "move" the treated/control to both have $p(X)$ instead of $p(X|D = 1)$ and $p(X|D = 0)$

What does it mean for weights to “move a distribution”?



Here we move a control distribution to look like treated (as you would for ATT), but the point is about the role of weights in shifting a distribution.

What weights achieve this?

For ATE, we wish we could average over $p(X)$ for treated and controls,

$$\int \mathbb{E}[Y_i|D_i = 1, X_i]p(X)dx - \int \mathbb{E}[Y_i|D_i = 0, X_i]p(X)dx$$

But nature gives us this DIM:

$$\int \mathbb{E}[Y_i|D_i = 1, X_i]p(X_i|D = 1)dx - \int \mathbb{E}[Y_i|D_i = 0, X_i]p(X_i|D_i = 0)dx$$

Consider weighting by $\frac{p(X_i)}{p(X_i|D_i)}$: (aka $\frac{p(D_i)}{p(D_i|X_i)}$)

$$\begin{aligned}
\mathbb{E}[\hat{\tau}_{IPW}] &= \int \mathbb{E}[Y_i|D_i = 1, X_i]\frac{p(X|D = 1)p(X_i)}{p(X_i|D_i = 1)}dx \\
&\quad - \int \mathbb{E}[Y_i|D_i = 0, X_i]\frac{p(X|D_i = 0)p(X_i)}{p(X_i|D_i = 0)}dx \\
&= \int \mathbb{E}[Y_i|D_i = 1, X_i]p(X_i)dx - \int \mathbb{E}[Y_i|D_i = 0, X_i]p(X_i)dx
\end{aligned}$$

The duality of weights

Those weights ($\frac{p(D_i)}{p(D_i|X_i)}$) are exactly the (stabilized) IPW weights!

That is, the weights that get you balance are also the weights you'd get with the (real) IPW.

This gives us two (flawed) ways forward

- try to compute the IPWs...but this requires a model, $\widehat{p(D_i|X_i)}$
- try to get balance on X...but you have to pick some specific functions of X to balance
- In other words: choosing weights by IPW and by balancing are the same in theory but will be different in practice with different weaknesses.

The first has a longer history and we've talked about it, let's get to the second...

Weighting for balance

In these approaches, forget about trying to model the treatment, just choose weights that give you "balance":

- Ideally you want to get $p(X|D = 1)$ to equal $p(X|D = 0)$.
- Often people settle for equal means alone, i.e. $\mathbb{E}[X|D = 0] \approx \mathbb{E}[X|D = 1]$.
- Though we can try to get equal means on X^2, $X_1 X_2$, etc. as well
- It turns out that these get you unbiased ATT estimates when Y_{0i} is linear in the things you get mean balance on

Weighting for covariate balance

To be explicit: find weights on control units s.t. the mean of X (or other functions of X) among controls matches that of treated.

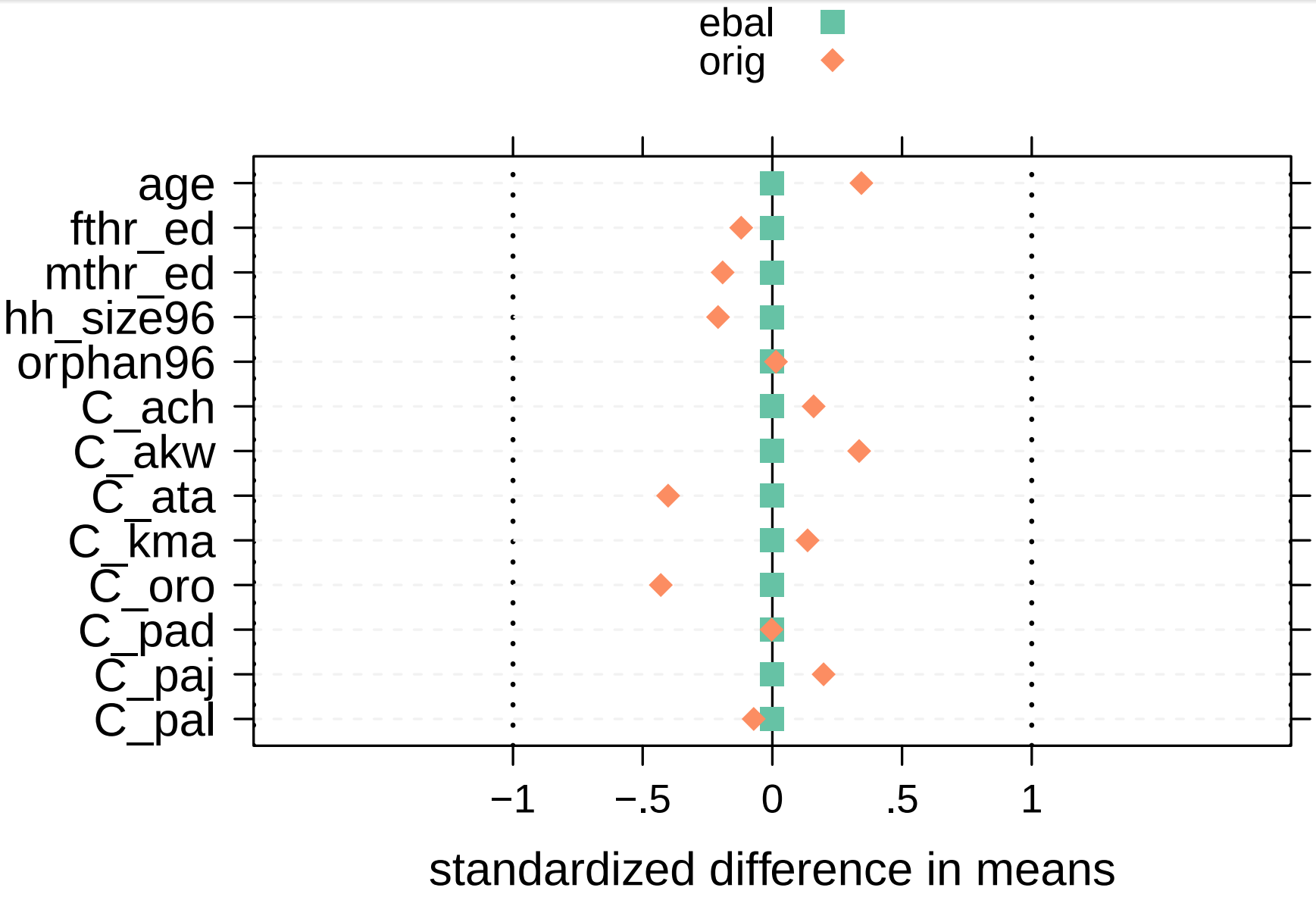
Resulting weights can be used to get weighted difference in means, or applied to any outcome model (e.g. regression)

- Could also move treated to controls (for what estimand?), or move both to overall means (for what estimand?)
- You can often find perfect mean balance, no iteration
- Standard errors are still an open question. Bootstrap likely okay.

Get equal means: entropy balancing (Hainmueller, 2011), `ebal`

To get multivariate (smoothed) balance, $\hat{p}(X|D=0) = \hat{p}(X|D=1)$, Kernel Balancing (`kbal`, see my website).

Balance on Blattman data with Ebal



Model-based Estimation of Causal Effects

Regression popular to "control for" variables. Does this allow causal inference?

Take the quantity:

$$\mathbb{E}[\tau(x)] = \mathbb{E}[Y_{1i} \mid X_i = x] - \mathbb{E}[Y_{0i} \mid X_i = x]$$

Which under CI is identifiable with estimator:

$$\mathbb{E}[\tau(x)] = \mathbb{E}[Y_i \mid D_i = 1, X_i = x] - \mathbb{E}[Y_i \mid D_i = 0, X_i = x]$$

A model-based approach for estimating causal effects:

- Suppose $\mathbb{E}[Y_{0i}|X_i] = \mathbb{E}[Y_i \mid D_i = 0, X_i] = \beta_0 + \gamma^\top X_i$
- ...and $\mathbb{E}[Y_{1i}|X_i] = \mathbb{E}[Y_i \mid D_i = 1, X_i] = \beta_0 + \beta_1 + \gamma^\top X_i$
- Which means $\beta_1 = \mathbb{E}[\tau(X)]$
- As single model:

$$\mathbb{E}[Y_i \mid D_i, X_i] \;=\; \beta_0 + \beta_1 D_i + \gamma^\top X_i,$$

OLS as an Estimator of Causal Effects

When is $\hat{\beta}_{OLS}$ a good estimator of τ_{ATE}?

We might posit two assumptions:

(1) Constant treatment effect. We assumed

- $\mathbb{E}[\tau(X)]$ is always one number...
- i.e., constant (expected) treatment effects (by strata of X)

(2) Linearity: By which we really mean

$$Y_{id} = \beta_0 + \beta_1 D_i + \gamma^\top X_i + \varepsilon_i \quad \text{with } \mathbb{E}[\varepsilon_i | X_i, D_i] = 0$$

Where $\mathbb{E}[\varepsilon_i | X_i, D_i] = 0$ (the "conditional independence assumption")
Noting that (2) implies (1) there are 3 possible scenarios:

1. Both (1) and (2) are true.
2. Only (1) is true.
3. Neither (1) nor (2) is true.

Case 1: Constant Effect w/ Linear Potential Outcomes

Result: If treatment effect is constant across units and potential outcomes are linear in X_i, then the OLS estimate of β_1 in the following regression model

$$Y_i = \beta_0 + \beta_1 D_i + \gamma^\top X_i + \varepsilon_i$$

is an unbiased and consistent estimator of τ_{ATE}.

Proof (just how we got here):

$$\begin{aligned}
\mathbb{E}[\beta_1] &= \mathbb{E}_x \mathbb{E}[Y_i | D_i = 1, X_i] - \mathbb{E}[Y_i | D_i = 0, X_i] \quad \text{(correct specification)} \\
&= \mathbb{E}_x \mathbb{E}[Y_{1i} | X_i] - \mathbb{E}[Y_{0i} | X_i] \quad \text{(SOO)} \\
&= \mathbb{E}_x \tau(X) \\
&= \tau \quad \text{(constant effect assumption)}
\end{aligned}$$

Note equivalence: CI+linearity $\rightarrow \varepsilon$ cannot be related to D given X (CIA):

$$\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp D_i \mid X_i \quad \Longrightarrow \quad \varepsilon_i \perp\!\!\!\perp D_i \mid X_i$$

Case 2: Constant Effect w/ Unknown Functional Form

Linearity is a very strong assumption in most practical situations.

What if $\mathbb{E}[Y_i(d)|X_i, D_i]$ is an unknown, nonlinear function of X and D but we use OLS?

Recall that OLS is still the best linear predictor in terms of MSE:

$$\hat{\beta}_{OLS} = \underset{\hat{\beta}_1}{\text{argmin}} \; \mathbb{E}\left[(Y_i - \hat{\beta}_0 - \hat{\beta}_1 D_i - \hat{\gamma}^\top X_i)^2\right]$$

Implies $\hat{\beta}_{OLS}$ is best linear approximation to the population regression function:

$$\hat{\beta}_{OLS} = \underset{\hat{\beta}_1}{\text{argmin}} \; \mathbb{E}\left[\left(\mathbb{E}[Y_i \mid D_i, X_i] - \hat{\beta}_0 - \hat{\beta}_1 D_i - \hat{\gamma}^\top X_i\right)^2\right]$$

- $\therefore$ we say "$\hat{\beta}_{OLS}$ is best linear approximation to the true treatment effect", whatever the true functional form is.
- This approximation may or may not be good! Danger of mis-specification bias.
- More flexible models (nonlinear, semi-/non-parametric, etc.) are a good idea

Case 3: Heterogeneous Treatment Effects

We typically cannot assume constant treatment effects. Is this a problem?

Recall the subclassification estimator for the ATE:

$$\hat{\tau}_{ATE} = \sum_{x \in \mathcal{X}} \{\mathbb{E}[Y_i \mid D_i = 1, X_i = x] - \mathbb{E}[Y_i \mid D_i = 0, X_i = x]\} \Pr(X_i = x)$$

Similarly, the subclassification estimator for the ATT:

$$\hat{\tau}_{ATT} = \sum_{x \in \mathcal{X}} \{\mathbb{E}[Y_i \mid D_i = 1, X_i = x] - \mathbb{E}[Y_i \mid D_i = 0, X_i = x]\} \Pr(X_i = x \mid D_i = 1)$$

Result: The OLS estimator can be written as a subclassification estimator, where the weights come from conditional variances of D_i in each subgroup:

$$\hat{\beta}_{OLS} = \sum_{x \in \mathcal{X}} \{\mathbb{E}[Y_i \mid D_i = 1, X_i = x] - \mathbb{E}[Y_i \mid D_i = 0, X_i = x]\} \frac{\mathrm{Var}(\hat{D}_i \mid X_i = x)\Pr(X_i = x)}{\sum_{x'} \mathrm{Var}(\hat{D}_i \mid X_i = x')\Pr(X_i = x')}$$

- Intuition: OLS "learns more" from strata of X where π_X closer to 0.50.
- think about what this means for your estimate, especially with lack of overlap
- This is from Angrist and Pischke. It depends on D being linear in X. For more general weights, see my Shinkre & Hazlett.

OLS as a Subclassification Estimator

Estimator	Weights for Subgroups	Unbiased for
$\hat{\tau}_{ATE}$	$\Pr(X_i = x)$	τ_{ATE}
$\hat{\tau}_{ATT}$	$\Pr(X_i = x \mid D_i = 1)$	τ_{ATT}
$\hat{\beta}_{OLS}$	$\dfrac{\mathrm{Var}(\hat{D}_i \mid X_i = x)\Pr(X_i = x)}{\sum_{x'} \mathrm{Var}(\hat{D}_i \mid X_i = x')\Pr(X_i = x')}$	"$\tau_{\text{CVW-ATE}}$"

- With heterogeneous treatment effects, OLS provides an unbiased estimator for a *conditional-variance-weighted average treatment effect*.
- Estimand has a causal meaning, but difficult to interpret. Not the ATE or ATT.
- Note: $\mathrm{Var}(\hat{D}_i \mid X_i = x) = \hat{\pi}(x)(1 - \hat{\pi}(x))$. Thus,
  - what happens when $\hat{\pi}_i$ is 0 or 1?
  - largest weights for strata $\hat{\pi}_i$ close to 0.5,
- This result assumes discrete Xs, but intuition holds for continuous Xs.

Improvements over pooled OLS

- **Imputation**: use regression to impute missing potential outcomes. E.g. Estimate $\mathbb{E}[Y_{0i}|X_i]$ and use as counterfactuals for observed Y_{1i}. Can weight each pairwise difference as you like. Aka g-computation, T-learner.
- **Interactive** (i.e. Lin). Include interaction of X with D to allow separate slope for $\mathbb{E}[Y(0)|X]$ and $\mathbb{E}[Y(1)|X]$.

It turns out these are precisely the same thing!

- Both relax your (modeling) assumption from "Y is linear in D and X" (which implies constant treatment effects) to "Y(1) is linear in X, and Y(0) is linear in X", which we call "separate linearity" (Shinkre & Hazlett)
- Unless you somehow care about $\tau_{CVW-ATE}$ as such, the only reason to use OLS without interaction/imputation is that it would be more efficient in the case where you really do have constant treatment effects.

Combining Regression, Matching and Weighting

The different logics of matching, weighting, and regression can be combined to improve finite-sample performance or robustness:

- Bias-corrected matching (Abadie and Imbens 2005):
  - estimate bias due to unequal Y_{0i} of matched units due to imperfect match on X's.
  - subtract it off from the matching estimate for correction
- "Doubly-robust" estimation by weighted regression:
  - Use weighting (e.g. IPW or balancing weights) in a regression that also includes covariates
- Construct estimators that employ both an outcome model and a weighting of residuals from it.
  - See especially double-ml (DML), augmented IPW (AIPW), targeted maximum likelihood (tMLE)

Summary: Estimation under Conditional Ignorability

- Matching, weighting and regression are used estimate average causal effects when one can assume conditional ignorability
- These are estimation strategies; the validity of the identification strategy (SOO) remains a first-order concern
- No single method is dominant at the moment, and the tides will change...
- My two cents: Ideally the estimator doesn't matter so much; best if you can use 5 different ones and get similar results. Differences in results often point to poor overlap/ model dependency problems that are worth exploring anyway.